

Matgeo Presentation - Problem 8.2.21

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October 12, 2025

Problem Statement

Find the equation of the conic that satisfies the given conditions:

- vertex $(0,0)$

- passing through $(5,2)$

- symmetric with respect to Y axis.

Solution: General Equation

The general conic equation is:

$$g(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1)$$

For a parabola with vertex at the origin and axis along the Y-axis, the focus is $\mathbf{F} = \begin{pmatrix} 0 \\ p \end{pmatrix}$ and the directrix is $y = -p$. This gives $\mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $c = -p$.

Solution: Deriving Parameters

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - \mathbf{n}\mathbf{n}^\top = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (2)$$

$$\mathbf{u} = c\mathbf{n} - \mathbf{F} = (-p) \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ p \end{pmatrix} = \begin{pmatrix} 0 \\ -2p \end{pmatrix} \quad (3)$$

$$f = \|\mathbf{F}\|^2 - c^2 = p^2 - (-p)^2 = 0 \quad (4)$$

Solution: Equation of Parabola

Substituting these into the general conic equation (1):

$$\mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} 0 & -2p \end{pmatrix}^\top \mathbf{x} + 0 = 0 \quad (5)$$

$$x^2 - 4py = 0 \quad (6)$$

The curve passes through the point (5,2).

$$(5)^2 - 4p(2) = 0 \quad (7)$$

$$25 - 8p = 0 \quad (8)$$

$$p = \frac{25}{8} \quad (9)$$

Solution: Final Equation

The final parameters are:

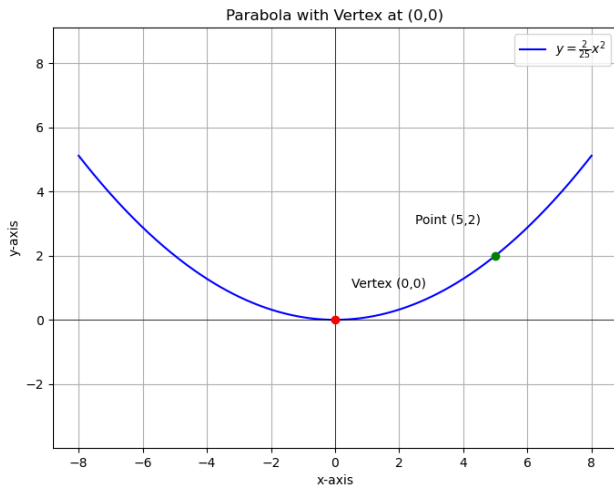
$$\mathbf{v} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ -\frac{25}{4} \end{pmatrix}, \quad f = 0 \quad (10)$$

The equation of the parabola is:

$$x^2 = 4 \left(\frac{25}{8} \right) y \quad (11)$$

$$x^2 = \frac{25}{2} y \quad (12)$$

Plot



C Code: parabola_calculator.c

```
double calculate_y(double x) {  
    double y = (2.0 / 25.0) * x * x;  
    return y;  
}
```


Python Code: main_caller.py

```
import ctypes
import os

lib_path = os.path.join(os.getcwd(), 'libparabola.so')
parabola_lib = ctypes.CDLL(lib_path)

parabola_lib.calculate_y.argtypes = [ctypes.c_double]
parabola_lib.calculate_y.restype = ctypes.c_double

x_test = 5.0
y_result = parabola_lib.calculate_y(x_test)

print(f"Calling C function with x={x_test}")
print(f"The calculated y value is: {y_result}")
```

Python Code: `plotter.py`

```
import numpy as np
import matplotlib.pyplot as plt

x_vals = np.linspace(-8, 8, 400)
y_vals = (2.0 / 25.0) * x_vals**2

plt.figure(figsize=(8, 6))

plt.plot(x_vals, y_vals, label=r'$y=\frac{2}{25}x^2$', color='blue')

plt.scatter([0], [0], color='red', zorder=5)
plt.annotate('Vertex (0,0)', xy=(0, 0), xytext=(0.5, 1))

plt.scatter([5], [2], color='green', zorder=5)
plt.annotate('Point (5,2)', xy=(5, 2), xytext=(2.5, 3))

plt.title('Parabola with Vertex at (0,0)')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
```